The Implied Cost of Capital: A New Approach

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Abstract

We propose a new approach to estimate the implied cost of capital (ICC). Our approach is distinct from the prior literature in that we do not rely on analysts' earnings forecasts to compute the ICC. Instead, we use a cross-sectional model to forecast the earnings of individual firms. Our approach has two major advantages. First, it allows us to estimate the ICC for a much larger sample of firms over a much longer time period. Second, it is not affected by the various issues that lead to the well-documented biases in analysts' forecasts. Our cross-sectional earnings model delivers earnings forecasts that outperform consensus analyst forecasts. We show that, as a result, our approach to estimate the ICC produces a more reliable proxy for expected returns than other approaches. We present evidence on the implications for the equity premium and a variety of asset pricing anomalies.

Keywords: Cost of capital, expected return, earnings forecasts, equity premium, anomalies
JEL Classification Codes: G12, G17, G31, M41.

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Estimating a firm's expected stock return (or cost of equity capital) is essential for testing the tradeoff between risk and return, a central theme in modern finance. Expected stock returns also play a key role in capital budgeting and other corporate finance decisions, and are an important input to investment management practice.

Prior academic studies almost exclusively rely on realized (ex post) stock returns to measure ex ante expected returns. However, as many researchers (e.g., Blume and Friend, 1973; Sharpe, 1978; Froot and Frankel, 1989; Elton, 1999) point out, realized returns are a very noisy proxy for expected returns. Empirically, expected return estimates that are based on realized returns have proven unsatisfactory in many regards. Elton (1999) provides examples demonstrating that realized returns can deviate significantly from expected returns over prolonged periods of time. Traditional asset pricing models such as the CAPM and the APT as well as empirically motivated models such as the Fama and French (1993) three-factor model also generate expected return estimates, but these are notoriously imprecise (Fama and French, 1997).

To address these deficiencies, recent accounting and finance studies (e.g., Claus and Thomas, 2001; Gebhardt, Lee, and Swaminathan, 2001; Pástor, Sinha, and Swaminathan, 2007) propose an alternative approach to estimate expected returns: the implied cost of capital (ICC).[1] The ICC of a given firm is the internal rate of return that equates the firm's stock price to the present value of all expected cash flows (measured by analysts' earnings forecasts). In other words, it is the discount rate that the market uses to discount the expected cash flows of the firm. The main advantage of the ICC approach is that it does not

[1] We refer to Easton (2008) for a comprehensive review of this literature.

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rely on noisy realized returns or on a specific asset pricing model. Instead, it derives expected returns from stock prices and earnings forecasts.

The idea behind the ICC is simple and intuitively appealing. As a result, the ICC has gained popularity in recent studies on empirical asset pricing (e.g., Chava and Purnanandam, 2007; Chen and Zhao, 2007; Pástor, Sinha, and Swaminathan, 2007) and on governance and disclosure (e.g., Francis, Khurana, and Pereira, 2005; Hail and Leuz, 2006ab; Hribar and Jenkins, 2004). However, there is evidence suggesting that the performance of the ICC as a proxy for expected returns is not very satisfactory. Several authors (e.g., Easton and Monahan, 2005; Guay, Kothari, and Shu, 2005) examine the relation between the ICC and realized returns.[2] The general conclusion is that the ICC is not a reliable proxy for expected returns. For example, Easton and Monahan (2005) find that ICC estimates are negatively correlated with realized returns after controlling for proxies for cash flow news and discount rate news. They attribute the lack of reliability of the ICC to the quality of analysts' earnings forecasts, and call for additional research on accounting-based expected return proxies.[3]

There are other concerns about the analyst-based ICC estimates. First, it is not clear that analysts' earnings forecasts truly reflect market expectations. Although analysts' forecasts are used by a variety of market participants, they also exhibit important biases. A large number of studies (e.g., Easton and Sommers, 2007) document that analysts' forecasts tend to be too optimistic. Analysts also overreact (underreact) to good (bad) prior earnings

[2] From a standard Campbell and Shiller (1988) decomposition, we know that, by definition, realized returns are equal to the sum of expected returns, shocks to cash flows, and shocks to discount rates. Therefore, a good expected return proxy should be positively correlated with realized returns after controlling for changes in expectations about future cash flows and future discount rates.

[3] Abarbanell and Bushee (1997) and Francis, Olsson, and Oswald (2000) find large valuation errors when analysts' forecasts are used in valuation models.

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news, consistent with incentive-based explanations of analyst optimism (e.g., Abarbanell and Bernard, 1992; Dugar and Nathan, 1995; Lin and McNichols, 1998). Second, the coverage of analysts' forecasts is limited. Many firms do not have analyst following at all. For firms with analyst following, earnings forecasts beyond the first year are often not available. The IBES analysts' forecasts data are heavily tilted toward big firms, especially in the early years (Hong and Stein, 1999) and financially distressed firms are underrepresented (La Porta, 1996; Deither, Malloy, and Scherbina, 2002). Also, analyst coverage by IBES only starts in 1980, but many applications (e.g., the equity premium and the inter-temporal risk-return relation) require a long time-series of expected return estimates.

To address these issues, we propose a new approach to estimate the ICC. Building on the work of Fama and French (2000, 2006), Hou and Robinson (2006), and Hou and van Dijk (2008), we use a cross-sectional model to forecast the earnings of individual firms. These studies show that cross-sectional models are remarkably powerful in capturing variation in future profitability across firms based on publicly available information at the time of the forecast. We then feed the model-based earnings forecasts into the discounted residual income model to estimate the ICC for a large cross-section of U.S. stocks.

A major advantage of our cross-sectional approach is that it provides statistical power without imposing strict data requirements. Our approach allows us to compute the ICC for any firm with public equity and information on a limited number of accounting variables at the time of the forecast. As a result, the cross-sectional coverage of the ICC estimates is much larger than in studies that use analysts' forecasts. In addition, we are able to estimate the ICC for earlier periods during which the IBES analyst data are not available.

A second important advantage of forecasting earnings using a cross-sectional model is that the forecasts are not affected by issues related to analysts' incentives, which cause

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analysts' forecasts to be biased. Rather, the model provides a parsimonious and unbiased way of capturing market expectations of future earnings based on a limited set of public information that is available to market participants ex ante.

We show that our cross-sectional earnings model is very powerful in capturing variation in earnings performance across firms. The adjusted R^2 of the models explaining one-, two-, and three-year ahead earnings is 87%, 81%, and 77%, respectively. More importantly, the model produces earnings forecasts that closely match the consensus analyst forecasts in terms of forecast accuracy, but exhibit much lower levels of forecast bias and much higher levels of earnings response coefficients.

Following Easton and Monahan (2005), we assess the reliability of the ICC by assessing the correlation with realized returns (after controlling for proxies for cash flow news and discount rate news). We show that our ICC estimates are significantly positively correlated with realized returns, while – in line with the findings of Easton and Monahan (2005) – the traditional ICC estimates based on analysts' forecasts are negatively correlated with realized returns. A simple trading strategy based on a long position in stocks with a high ICC and a short position in stocks with a low ICC produces positive returns of close to 9% per annum for our model-based ICC estimates and negative returns of up to -10% for the analyst-based ICC estimates.

We provide evidence suggesting that the greater reliability of our ICC estimates stems from the superior earnings forecasts produced by the cross-sectional model. We sort stocks into tercile portfolios based on their forecast bias and accuracy as well as the firm-specific earnings response coefficient associated with the earnings forecasts. We then compute the correlation between the ICC and the realized stock returns within each of the

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tercile portfolios. We find that this correlation is highest for the tercile portfolios of stocks with the lowest forecast bias, the highest accuracy, and the highest earnings response coefficient. The correlations decrease as the bias increases and as the accuracy and the earnings response coefficient deteriorate. Hence, better earnings forecasts translate into a closer association between the ICC and realized stock returns.

Our new approach to estimate the ICC has important implications for a number of key issues in asset pricing. We reassess the equity premium and a variety of asset pricing anomalies based on our ICC estimates. Our analysis indicates that using our new, ex ante measure of expected returns leads to different inferences about the equity premium, the size effect, the accrual anomaly, the asset growth anomaly, and the analysts dispersion anomaly than studies relying on ex post measures of expected returns.

This study adds to recent papers that construct alternative proxies for expected stock returns, including Brav, Lehavy, and Michaely (2005), Campello, Chen, and Zhang (2008), and Hou and van Dijk (2008). Our more reliable estimate of a firm's ICC is not only useful for academic researchers, but also for corporate managers and for investment professionals interested in portfolio allocation, performance evaluation, and active risk management.

1. Data and empirical methodology

Our sample includes all NYSE, Amex, and Nasdaq listed securities with sharecodes 10 or 11 (i.e., excluding ADRs, closed-end funds, and REITs) that are contained in the intersection of the CRSP monthly returns file and the Compustat industrial annual file between July 1963 and December 2005. We use the following variable definitions. Earnings is net income before extraordinary items from Compustat. The market value of a

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firm is defined as its total assets plus Compustat market equity (stock price times the number of shares outstanding at fiscal year end) minus book equity. Book equity is Compustat stockholder's equity. Total assets and dividends are also from Compustat. We calculate operating accruals using the indirect balance sheet method as the change in non-cash current assets less the change in current liabilities excluding the change in short-term debt and the change in taxes payable minus depreciation and amortization expense, as in Sloan (1996).

To forecast earnings at the firm level, we estimate the following pooled cross-sectional regression for each year t between 1967 and 2005, using the previous ten years (three years minimum) of data:

$$E_{i,t+\tau} = \alpha_0 + \alpha_1 V_{i,t} + \alpha_2 A_{i,t} + \alpha_3 D_{i,t} + \alpha_4 DD_{i,t} + \alpha_5 E_{i,t} + \alpha_6 Neg\,E_{i,t} + \alpha_7 AC_{i,t} + \varepsilon_{i,t+\tau}, \quad (1)$$

where $E_{i,t+\tau}$ (τ = 1, 2, or 3) denotes the earnings of firm i in year $t+\tau$ and all explanatory variables are measured at the end of year t: $V_{i,t}$ is the market value of the firm, $A_{i,t}$ is the total assets, $D_{i,t}$ is the dividend payment, $DD_{i,t}$ is a dummy variable that equals 0 for dividend payers and 1 for non-payers, $Neg\ E_{i,t}$ is a dummy variable that equals 1 for firms with negative earnings (0 otherwise), and $AC_{i,t}$ is the operating accruals. To mitigate the impact of extreme observations on the regression coefficients, we winsorize each variable annually at the 0.5% and 99.5% percentiles.

Equation (1) is a variant of the cross-sectional profitability models used in Fama and French (2000, 2006), Hou and Robinson (2006), and Hou and van Dijk (2008). These authors demonstrate that cross-sectional models are able to explain a substantial fraction of the variation in future profitability across firms based on publicly available information that is known ex ante. The main differences between our model and theirs are that we use

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pooled regressions to forecast the levels of earnings for the next three years, whereas they use annual cross-sectional regressions to forecast profitability (earnings over lagged assets) for the next year. We forecast levels of earnings instead of profitability to make our forecasts more consistent with analysts' forecasts, which concern the level of earnings per share. (We obtain similar results when we estimate a model to forecast earnings over lagged assets (ROA) instead of the level of earnings.)

For each firm, we estimate expected earnings for years t+1, t+2, and t+3 (i.e., $E_t[E_{t+1}]$, $E_t[E_{t+2}]$, and $E_t[E_{t+3}]$, where $E_t[]$ is the expectation operator) by multiplying the independent variables observed at the end of year t with the coefficients from the pooled regression estimated using the previous ten years (three years minimum) of data. This is to ensure that our earnings forecasts are out of sample (that is, all the information that is required to forecast earnings for years t+1, t+2, and t+3 is available at the end of year t).

The ICC is the internal rate of return that equates the current stock price to the present value of expected future cash flows. One common approach is to estimate the ICC using the discounted residual income model, which has the following general form:

$$P_{i,t} = BPS_{i,t} + \sum_{k=1}^{\infty} \frac{E_t[(ROE_{i,t+k} - R_i) \times BPS_{i,t+k-1}]}{(1+R_i)^k}, \tag{2}$$

where $P_{i,t}$ is the stock price of firm i, $BPS_{i,t}$ is book equity per share, $E_t[]$ denotes market expectations formed at the end of year t, and $(ROE_{i,t+k} - R_i) \times BPS_{i,t+k-1}$ is the firm's residual income for year t+k, defined as the difference between the after tax return on book equity and the cost of equity capital multiplied by book equity per share at the end of the previous year. Intuitively, a firm's residual income measures its ability to earn income beyond that required by equity investors. Assuming "clean surplus" accounting, equation (2)

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is equivalent to the familiar dividend discount model.[4] Accounting studies (e.g., Penman and Sougiannis, 1998; Francis, Olsson, and Oswald, 2000; Gebhardt et al., 2001) argue that the residual income model is better at capturing the effect of economic profits on firm value, and the resulting valuation is less sensitive to assumptions about long-term growth rates.

We estimate the ICC as the positive cost of equity capital R_i that solves an adapted version of equation (2):

$$M_{i,t} = B_{i,t} + \sum_{\kappa=1}^{11} \frac{\mathrm{E}_t[(ROE_{i,t+\kappa} - R_i) \times B_{i,t+\kappa-1}]}{(1+R_i)^{\kappa}} + \frac{\mathrm{E}_t[(ROE_{i,t+12} - R_i) \times B_{i,t+11}]}{R_i \times (1+R_i)^{11}}. \qquad (3)$$

This equation is identical to the model of Gebhardt et al. (2001), but expresses firm valuation in terms of market equity ($M_{i,t}$) and book equity ($B_{i,t}$) instead of stock price and book equity per share. We estimate expected *ROE* for the next three years using the earnings forecasts from our cross-sectional model and book equity determined based on clean surplus accounting and a constant dividend payout ratio ($B_{i,t+\tau} = B_{i,t+\tau-1} + \mathrm{E}_t[E_{i,t+\tau}] - \mathrm{E}_t[D_{i,t+\tau}]$, where $\mathrm{E}_t[D_{i,t+\tau}]$ is the forecasted dividend for year $t+\tau$, estimated using the current dividend payout ratio; for firms experiencing negative earnings, we divide the dividends paid by (0.06* total assets) to derive an estimate of the payout ratio, following Gebhardt et al., 2001). After year t+3, we assume that the *ROE* mean-reverts to the historical industry median value by year t+11; after that the residual income becomes a perpetuity.[5]

We estimate the ICC for each firm at the end of June of each calendar year t using the end-of-June market value and the earnings forecasts as of the most recent fiscal year end. We then match the ICC estimates with annual realized stock returns compounded from July of year t to June of year t+1.

[4] Clean surplus accounting requires that all gains and losses affecting book equity are included in earnings. In other words, the change in book equity is equal to earnings minus net dividends.

[5] Following Gebhardt et al. (2001), we exclude loss firms when calculating the industry median ROE.

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2. Properties of the cross-sectional earnings model

Table 1 shows the average coefficients from the pooled cross-sectional regressions to explain earnings in years t+1, t+2, and t+3, as well as their time-series t-statistics. The average coefficients for most of the explanatory variables (except the negative earnings dummy) have the same sign for the one-, two-, and three-year ahead earnings regressions. Consistent with the results of Fama and French (2006), Hou and Robinson (2006), and Hou and van Dijk (2008), earnings are highly persistent. The coefficients on year t earnings are 0.6839, 0.5981, and 0.5398 and highly statistically significant for the one-, two-, and three-year ahead earnings regressions, respectively.[6] Earnings are negatively related to total assets and positively to the market value of the firm. Firms that pay out more dividends and firms with lower operating accruals have higher future earnings. As expected, the dummy variable for non-dividend payers has a positive coefficient. The coefficient on the negative earnings dummy is negative for year t+1, and positive for years t+2 and t+3, although it is only statistically significant in the three-year ahead regression.

Our model captures a very large part of the variation in future earnings across firms using variables that are strictly ex ante. The average regression R^2 is 87% for the one-year ahead earnings regression, 81% (two-year ahead), and 77% (three-year ahead). This performance is remarkable considering the parsimonious specification of equation (1).

Table 2 reports the average one-, two-, and, three-year ahead value-weighted average earnings forecasts produced by our cross-sectional model for each year in the sample period as well as the number of firms for which the model-based earnings forecasts

[6] Since we estimate the model every year over the previous ten years of data, the t-statistics in Table 1 are biased. Unreported results show that many of the t-statistics are still significant after correcting for overlapping data. We note that we only use the coefficients, and not the t-statistics, to compute the earnings forecasts.

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are available in that year. The forecasts start in 1967, because we require at least three years of data to estimate the cross-sectional earnings regression. Table 2 also shows the value-weighted average consensus analyst earnings forecasts for each year and the number of firms for which the analyst-based earnings forecasts are available. To facilitate comparison, we express both the model-based and the analyst-based earnings forecasts as a fraction of the firm's market capitalization and stock price, respectively, as of end of June of each year. That is, we divide the earnings forecasts produced by the model by the market capitalization and the analysts' forecasts (which are on a per share basis) by the share price.

The number of firms for which the model yields earnings forecasts grows from slightly above 1,000 in the early years of the sample period to almost 4,000 in the late 1990s, after which it drops to around 3,000 in the most recent years. We observe a decline in average earnings forecasts since the late 1970s, consistent with the evidence of Fama and French (2004) that U.S. publicly traded firms have become less profitable over time. The average analyst-based forecasts exhibit a similar time-series pattern.

The difference in the coverage of the model-based earnings forecasts and analyst-based forecasts is striking. In the period 1965-1979, IBES data are not available at all. Starting in 1980, we have data on one- and two-year ahead analysts' earnings forecasts, but the initial coverage is limited, reaching levels that are comparable to those of the model-based forecasts only by the mid-1990s. We can only obtain three-year ahead analysts' earnings forecast starting in 1983, and only for a substantially smaller number of firms when compared to the model. Even as recently as 2005, three-year ahead earnings forecasts based on IBES data are available for just 1,721 firms (not tabulated), while the model allows us to construct earnings forecasts for 2,890 firms. We follow prior studies in the ICC

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literature and estimate an imputed three-year ahead forecast from the consensus long-term growth forecast and the two-year ahead earnings forecast for firms for which the three-year ahead IBES forecast is not available. This treatment boosts the number of firms with available three-year ahead forecasts based on IBES data for 2005 from 1,721 to 2,755 and the total number of firm-year observations for the three-year ahead forecasts from 16,046 to 55,824. We note that less than one third of these firm-year observations are thus actually based on three-year ahead earnings forecasts supplied by analysts.

The differences in coverage imply that we are able to estimate the model-based ICC for a much larger sample of firm-years than studies that rely on IBES data. Not only does the greater number of observations enhance the power of any asset pricing tests performed using the ICC, but it also allows us to address asset pricing issues that require a long time-series of expected return estimates and/or require expected return estimates for small or distressed firms, for which analysts' forecasts are scarce.

3. Properties of the earnings forecasts generated by the cross-sectional model

Table 3 compares the forecast bias, the forecast accuracy, and the earnings response coefficient (ERC) of the earnings forecasts generated by our cross-sectional earnings model with those of the analysts' forecasts. The table presents the time-series averages of the value-weighted bias and accuracy and of the cross-sectional ERC of the one-, two-, and three-year ahead earnings forecasts. Panel A reports the bias, accuracy, and ERC of the model-based forecasts for the full sample of firm-year observations for which these forecasts are available over the period 1967-2005. To compare our model-based forecasts with the analysts' forecasts, Panels B and C report the bias, accuracy, and ERC of the

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model-based and the analysts' forecasts for the sample of firm-years for which both forecasts are available (the "common sample") , which limits the sample period to 1980-2005.

The forecast bias is defined as the difference between the realized earnings and the earnings forecast scaled by market equity for model-based forecasts or by price for analysts' forecasts.[7] A negative bias means that the forecast is too optimistic. The results for the common sample indicate that the average bias of the forecasts based on the cross-sectional earnings model is substantially smaller than the bias of the analysts' forecasts at all three forecasting horizons. Consistent with prior studies, we find that analysts tend to be overly optimistic. The analyst bias is negative for all horizons, with the magnitude of the bias increasing with the forecast horizon. The cross-sectional earnings model also overestimates future earnings on average, but the average bias is much smaller when compared to the analysts' forecasts.

It is important to note that the three-year ahead forecasts have by far the biggest impact on the estimates of a firm's ICC, since year t+3 earnings play a dominant role in estimating the firm's expected profitability from year t+4 and on (see Section 1). For the year t+3 earnings forecasts, the difference between the bias of the model-based and the analysts' forecasts is particularly pronounced. The average forecast bias is -0.0076 for the model-based earnings forecasts and -0.0189 for the analysts' forecasts.

For the forecast accuracy, the picture is more balanced. The forecast accuracy is the absolute value of the forecast bias. A lower number means a more accurate earnings forecast. Overall, the differences between the accuracy of the model-based forecasts and the

[7] We measure realized earnings using net income before extraordinary items from Compustat for model-based forecasts and actual earnings per share provided by IBES for analysts' forecasts.

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analysts' forecasts are fairly small for all forecast horizons, although the difference is growing with the forecast horizon. But even for the three-year ahead forecasts where analysts' forecasts are most accurate relative to the model-based forecasts, the difference between the two is considerably smaller than that of forecast bias.

The earnings response coefficient (ERC) offers another metric to judge the performance of the model-based forecasts relative to the analysts' forecasts. The ERC captures the sensitivity of stock returns to unexpected earnings (i.e., the difference between realized and forecasted earnings). This type of analysis dates back to Ball and Brown (1968) and Lev and Ohlson (1982) and provides additional insights into the reliability of earnings forecasts. A strong stock market response to unexpected earnings indicates that the underlying earnings forecasts do a good job in capturing market expectations about future profitability.[8]

We estimate the ERCs for the three forecast horizons by running annual cross-sectional regressions of one-, two-, and three-year ahead realized returns on forecast biases measured over the same horizon. The forecast biases are standardized to have unit variance to make the ERCs comparable between model-based and analysts' forecasts.

The ERCs of the model-based forecasts for the common sample are equal to 0.0878, 0.4166, and 0.9639 for the one-, two-, and three-year forecast horizons. Unreported results show that all ERCs are highly statistically significant. The ERCs of the analysts' forecasts are 0.1066, 0.1935, and 0.4127. The ERC of the model-based forecasts is thus slightly smaller than the ERC of the analysts' forecasts at the one-year forecast horizon, but much

[8] The ERC is arguably the most direct way to evaluate whether earnings forecasts reflect market expectations. See Brown (1993), O'Brien (1988), and Wiedman (1996), among others.

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greater at the two- and three-year horizons.[9] The stock market thus reacts more strongly to earnings surprises relative to the model-based earnings forecasts than relative to the analysts' forecasts. This finding suggests that the model-based forecasts are a better measure of market expectations.

The model-based earnings forecasts for the full sample on average have a lower forecast accuracy and a smaller ERC than for the common sample. This makes sense, as the firms for which IBES data are not available tend to be smaller, more volatile firms for which earnings are harder to predict and stock returns are more noisy. The average forecast bias of the model-based forecasts is very close to zero for the full sample. Our parsimonious cross-sectional model thus on average produces unbiased earnings forecasts for a large sample of firms over an extended period of time.

4. Properties of the model-based ICC

So far, we have shown that the cross-sectional earnings model is remarkably powerful in explaining differences in future earnings across firms. The earnings forecasts produced by the model closely match analysts' forecast in terms of forecast accuracy, but are superior in terms of forecast bias and earnings response coefficient – especially for the crucial three-year horizon. In this section, we examine the attributes of the ICC estimates using the earnings forecasts generated by the model and compare these to the analyst-based ICC.

Table 4 presents summary statistics of the ICC estimates for each year in the sample period. The table shows the number of firms for which we are able to estimate the ICC as

[9] As discussed before, the long-run forecasts have the largest effect on the estimated ICC in the residual income model.

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well as the mean, standard deviation, and the 25th, 50th (median), and 75th percentiles of the annual distribution of ICC estimates across firms.

The coverage of the model-based ICC estimates is around 1,000 firms in the first few years, but rapidly increases to more than 2,500 from 1975 onwards. In total, we are able to obtain an estimate of the ICC for 102,067 firm-years. Averaged over 1968-2006, the mean model-based ICC is 13.28% and the median is 11.21%. As we should expect, there is considerable variation in the ICC estimates across firms.

The coverage of the analyst-based ICC starts in 1982 and is much more limited than the coverage of the model-based ICC in the 1980s and the 1990s. Only in recent years does the coverage of both approaches converge. The total number of firm-year observations in the analyst-based ICC sample is 51,572. Averaged over 1982-2006, the mean and median analyst-based ICC are 10.29% and 9.43%. (The lower estimates compared to the model-based ICC can be explained by the decline in the cost of capital during the 1980s and 1990s, see the discussion in Section 5.1 below.) Figure 1 plots the equal-weighted (Panel A) and value-weighted (Panel B) average market-wide ICC. The ICC estimates are much smoother than realized returns, consistent with the intuition that expected returns only change slowly over time, while realized returns are very noisy.

Easton and Monahan (2005) criticize proxies for a firm's ICC based on analysts' forecasts because of their negative correlation with realized stock returns. Panel A of Table 5 reports the time-series averages of the cross-sectional correlations of the model-based as well as the analyst-based ICC estimates with realized returns. We calculate the correlations of the ICC estimates not only with the realized returns in the year following the computation of the ICC (denoted by r_{t+1}), but also with realized returns in the second and

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third year (denoted by r_{t+2} and r_{t+3}). Since the ICC is a weighted average of the discount rates for all future horizons implied by market prices, and since the ICC only changes slowly over time, a reliable estimate of the ICC should be positively correlated with realized returns in subsequent years as well.

Consistent with Easton and Monahan (2005), we find negative correlations of the analyst-based ICC estimates with realized returns. The average correlations are -0.0130 (*t*-stat = -0.54) for year *t*+1 and -0.0640 (*t*-stat = -2.77) and -0.0750 (*t*-stat = -3.80) for years *t*+2 and *t*+3, respectively. For the model-based ICC, we find a correlation of 0.0550 (*t*-stat = 2.83) with realized returns in year *t*+1 and of 0.0512 (*t*-stat = 3.03) and 0.0429 (*t*-stat = 2.65) with realized returns in years *t*+2 and *t*+3, respectively. These findings suggest that our new model-based ICC measure is a more reliable proxy for expected returns than the analyst-based ICC.

We note that a correlation of 0.0550 may seem modest. However, this finding is consistent with studies that apply the Campbell and Shiller (1988) decomposition to individual stock returns (e.g., Vuolteenaho, 2002; Campbell, Polk, and Vuolteenaho, 2009), which find that only a very small fraction of the variation in realized returns can be explained by variation in expected returns. In addition, below we report considerably higher correlations among certain subgroups of stocks. The important finding is that, in contrast to analyst-based ICC estimates, our model-based ICC estimates are significantly positively correlated with realized returns.

We carry out additional tests by running annual Fama-MacBeth (1973) cross-sectional regressions of realized returns on the model-based or the analyst-based ICC. Following Easton and Monahan (2005), we also run regressions that include proxies for

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cash flow news and discount rate news. We use their measure of cash flow news, which represents the changes in the earnings forecasts for the three horizons we consider. We compute discount rate news as the change in the ICC estimate for a given year multiplied by negative one. Since neither the cash flow news nor the discount rate news over a certain year should be predictable based on the ICC that is available at the beginning of the year, we orthogonalize the cash flow and discount rate news proxies with respect to the ICC.

Panel A of Table 5 presents the average coefficients from the regressions (and their time-series *t*-statistics). The coefficient on the model-based ICC is equal to 0.1318 (*t*-stat = 2.41) when the dependent variables is r_{t+1}. Realized returns are thus positively and significantly related to our model-based ICC estimates. We also observe this significantly positive relation for r_{t+2} and r_{t+3} and after controlling for cash flow and discount rate news. We note that the number of observations is smaller for the regression specifications that include the cash flow and discount rate news proxies, because computing the cash flow news proxy requires information on future realized earnings.

Since our proxies for cash flow and discount rate news are based on our estimates of market expectations of future earnings and our ICC estimates, the sign and significance of the coefficients on the cash flow and discount rate news proxies in the regressions provide further evidence on the reliability of our approach. As expected, the coefficients on both news proxies are positive and statistically significant.

In contrast to our results for the model-based ICC, we find a negative regression relation between the ICC based on analysts' forecasts and realized stock returns. The estimated coefficients vary between -0.0320 and -0.5315 across the different model specifications and horizons and are statistically significant in some specifications. We

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conclude from the differences in correlations and regression coefficients that our new approach to estimate the ICC yields a more reliable measure of expected returns than the approach based on analysts' forecasts.

To gauge the economic significance of the relation between ICC estimates and realized returns, in Panel B of Table 5 we sort all stocks into decile portfolios based on their estimated ICC at the end of June of each year and compute the equal-weighted and value-weighted average realized returns of each portfolio from July to June of next year. We also examine the realized returns of each portfolio in the second and third year after portfolio formation.

The results in Panel B show that both the equal-weighted and the value-weighted average realized (ex post) returns increase with the ex ante expected return measured by the model-based ICC. A zero investment portfolio that is long in stocks with a high model-based ICC and short in stocks with a low ICC yields an equal-weighted (value-weighted) average return of 8.88% (8.28%) per annum with a *t*-statistic of 2.43 (2.50) in the first year following portfolio formation. High model-based ICC stocks continue to significantly outperform low ICC stocks in the second and third year after the portfolio formation. The value-weighted average second- and third-year realized returns on the long-short portfolio are 7.57% and 5.92%, while the equal-weighted average returns remain close to 9% per annum.

The contrast with the analyst-based ICC is, again, striking. Sorting on analyst-based ICC produces a zero or negative but insignificant spread in realized returns in the first year following portfolio formation. The average return on the long-short portfolio is 0.32% (equal-weighted, *t*-stat = 0.08) and -4.57% (value-weighted, *t*-stat = -1.15). In subsequent

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years, the relation actually turns significantly negative. In the second and third years after portfolio formation, the long-short portfolio based on the analyst-based ICC shows negative returns that lie between -5.44% and -10.70%, with *t*-statistics ranging from -1.57 to -2.82.

This difference is not driven by the different sample period. Unreported results show that the portfolio sorts based on the model-based ICC over the period for which IBES data are available (after 1980) exhibit a pattern that is similar to the full sample period results. Over this period, the equal-weighted and value-weighted average realized returns on the long-short model-based ICC portfolio in the year following portfolio formation are 8.71% and 6.91%, respectively.

A potential explanation for the negative relation between the analyst-based ICC and realized returns is that the portfolio of firms with a high analyst-based ICC may to a disproportional extent consist of firms with a large negative forecast bias. These overly optimistic analysts' forecasts result in upwardly biased ICC estimates. If the market does not recognize the extent of the bias in analysts' forecasts, these firms are likely to experience negative cash flow shocks during subsequent years, depressing the realized returns.

The results in this section raise the question why the model-based ICC estimates perform better than the analyst-based estimates. Easton and Monahan (2005) and Easton and Sommers (2007), among others, argue that there is a link between the poor performance of ICC estimates based on analysts' forecasts and the poor quality of these forecasts. Table 6 sheds light on whether the greater reliability of our ICC estimates stems from the superior quality of the earnings forecasts delivered by our cross-sectional model. At the end of June of each year, we sort all stocks into tercile portfolios based on their

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forecast bias (Panel A), forecast accuracy (Panel B), or the earnings response coefficient (Panel C) associated with their model-based earnings forecasts for the next three years.[10] We then compute the cross-sectional correlation between realized returns and the model-based ICC estimates. To facilitate comparison with the results in Table 5, we also run annual Fama-MacBeth cross-sectional regressions of realized returns on the model-based ICC. Table 6 reports the average correlations as well as the average coefficients from the cross-sectional regressions and their time-series *t*-statistics within each of the tercile portfolios.

The relation between realized returns and the ICC estimates is strongest for the portfolios with the smallest forecast bias. The middle portfolio – which has an average forecast bias that is close zero – exhibits correlations between 0.1270 and 0.2009 for the three forecast horizons. The correlations decrease considerably as the forecast bias increases. The regression coefficients and their *t*-statistics confirm results based on the correlations. The coefficients are large, positive, and statistically significant for the middle portfolio, and the magnitude and the statistical significance of the coefficients are substantially smaller for tercile portfolios 1 and 3.[11]

The portfolios sorted on forecast accuracy also exhibit a clear pattern. The relation between realized returns and the ICC estimates is strongest for the firms with the most accurate earnings forecasts. The correlations decrease monotonically as the accuracy deteriorates. The magnitude of the regression coefficients and their *t*-statistics are also smaller for the portfolios of firms with less accurate forecasts.

[10] In the spirit of the cross-sectional ERC reported in Table 3, we estimate a firm-specific ERC for each firm in each year by dividing its one-, two-, and three-year ahead realized returns by the forecast biases measured over the same horizon. Firms with negative ERCs are excluded from the analysis.

[11] Unreported results show that the analyst-based ICC estimates are also positively correlated with realized returns for the tercile of firms with the smallest forecast bias.

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The correlation between the model-based ICC and realized returns also increases with the firm-specific ERC of the underlying earnings forecasts. For the low ERC tercile portfolio, the correlation is negative. But for the tercile portfolio with the greatest ERC, the correlation is positive and significant, ranging from 0.1277 to 0.1637 for the three forecast horizons.[12]

Overall, the results in Table 6 are supportive of the conjecture that the improved reliability of our ICC estimates stems from the greater quality of the underlying earnings forecasts.

5. Implications for asset pricing

Our new measure of expected stock returns – the ICC based on earnings forecasts generated by the cross-sectional earnings model – allows us to re-evaluate a wide variety of issues in empirical asset pricing. In this section, we present evidence on the equity premium and a number of asset pricing anomalies using our improved ICC measure. We note that many of these issues cannot be investigated using the analyst-based ICC estimates due to the limited number of firms and the limited sample period for which these estimates are available.

5.1 The equity premium

Table 7 shows estimates of the market equity premium in expected returns implied by the ICC estimates. For comparison, we also report the realized market equity premium based on the CRSP market index for NYSE/Amex/Nasdaq firms. Equal-weighted (value-

[12] Firms with forecast biases close to zero could produce extreme observations for our firm-specific ERC measure. To ensure that these extreme observations do not have undue influence on our results, we also winsorize the biases that are close to zero and recalculate the ERC. We find the results to be nearly identical to those reported in Panel C. For brevity, they are not tabulated.

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weighted) average excess returns are reported in Panel A (B). We use two different proxies for the risk-free rate: the annualized 30-day T-Bill rate and the 10-year Treasury constant maturity rate.

We find that the average realized equity premium is substantially greater than the ex ante equity premium implied by the ICC, consistent with Claus and Thomas (2001) and Fama and French (2002). The equal-weighted average realized excess market return over the period 1968-2006 is 9.60% relative to the T-Bill and 8.04% relative to the T-Bond rate. Our ICC estimates suggest that the equal-weighted expected equity premium is equal to 7.25% relative to the T-Bill and 5.69% relative to the T-Bond rate. The value-weighted equity premium in realized returns is 5.99% relative to the T-Bill and 4.43% relative to the T-Bond rate. The value-weighted expected equity premium implied by the ICC relative to the T-Bill (T-Bond) rate is only 2.60% (1.03%). We note that the choice of the risk-free rate has a substantial impact of the equity premium estimates.

Our estimate of the equity premium in expected returns is lower than estimates obtained by studies using analyst-based ICCs. Claus and Thomas (2001) report an equity premium of around three percent. Gebhardt et al. (2001) estimate the equity premium in expected returns to be 2.7%. Both of these papers use the 10-year rate as the risk-free rate. We find a value-weighted average equity premium of only 1.03% relative to the 10-year rate. One potential explanation for the differences between our numbers and the analyst-based ICC estimates is that the optimism in analysts' earnings forecasts leads to upwardly biased estimates of the ICC (e.g., Easton and Sommers, 2007).

In Mehra and Prescott's (1985) paper on the equity premium puzzle, they demonstrate that the variance-covariance matrix of aggregate consumption and stock and

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bond returns, when combined with a reasonable level of risk aversion, implies an equity premium slightly below one percent. The equity premium implicit in our model-based ICC estimates is consistent with this estimate.

Table 7 also displays estimates of the equal-weighted (Panel A) and value-weighted (Panel B) average realized and expected excess return for different industries. We classify firms into twelve industries based on their SIC codes using the definitions downloaded from Ken French's Web site. The equal-weighted (value-weighted) average realized excess return of the industry portfolios ranges from 8.33% (4.76%) to 13.67% (8.96%) relative to the T-Bill rate and from 6.77% (3.19%) to 12.10% (7.39%) relative to the T-Bond rate. The risk premia implied by our ICC estimates are considerably smaller. The industries with the highest value-weighted implied risk premium are Utilities (with a risk premium of 5.57% relative to the T-Bill and 4.01% relative to the T-Bond rate) and Consumer Durables (with a risk premium of 3.95% relative to the T-Bill and 2.38% relative to the T-Bond rate).

Figure 1 shows the time-series variation of the equal-weighted (Panel A) and value-weighted (Panel B) average market-level ICC as well as the equity premium implied by the ICC estimates relative to the T-Bill and T-Bond rates. Fama and French (2002) argue that their finding that realized market returns are higher than expected returns over the period 1951-2000 can be attributed to a decline in discount rates that generates large unexpected capital gains. The market-level ICC does show a clear and gradual decline since the early 1980s. The equal-weighted average ICC decreases from 20.21% in 1981 to 7.30% in 2005 and the value-weighted ICC declines from 14.34% in 1981 to 6.66% in 2005.[13] It is also

[13] The starting point of this significant decline coincides with the dramatic increase in the number of newly listed firms on major U.S. exchanges since the early 1980s. Fama and French (2004) raise the possibility that this increase is due to a decline in the cost of equity capital around the early 1980s. Our results support this explanation.

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interesting to note that the development of the implied risk premia over time suggests that the high stock market valuations in the late 1990s were driven by very low equity risk premia demanded by investors.

5.2 Anomalies

We investigate whether well-known anomalies that have been identified on the basis of realized stock returns also exist in ex ante expected returns as measured by our ICC estimates. We carry out univariate sorts on a number of individual firm characteristics that previous studies have shown to predict realized stock returns. We examine the following variables: beta, size, book-to-market equity (BE/ME), leverage, distress (e.g., Campbell, Hilscher, and Szilagyi, 2008), capital expenditures (CAPEX; Titman, Wei, and Xie, 2004), asset growth (Cooper, Gulen, and Schill, 2008), accruals (Sloan, 1996), net operating assets (NOA; Hirshleifer, Hou, Teoh, and Zhang, 2008), and dispersion in analysts' earnings forecasts (Diether, Malloy, and Scherbina, 2002).

Table 8 shows the equal-weighted (Panel A) and value-weighted (Panel B) average realized returns and ICC of decile portfolios sorted on these variables and the differences in the realized returns and the ICC between Decile 10 ("high") and Decile 1 ("low") as well as the *t*-statistic of the 10-1 spread.

The relation between beta and the ICC is similar to the relation between beta and realized returns. Both expected returns and realized returns are slightly lower for high beta portfolios. The difference between Decile 10 and Decile 1 is not statistically significant for realized returns, but it is significant for the ICC. In general, the *t*-statistics are much higher for the ICC estimates, since the ICC is much smoother than realized returns. The economic

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significance of the relation between beta and ICC is modest, with a value-weighted average 10-1 spread of -1.30% per annum.

The size effect is considerably stronger in expected returns than in realized returns. The average difference in realized returns between small and large firms is 5.07% (equal-weighted) and 4.44% (value-weighted), neither of which is statistically significant. However, we find an economically and statistically significant size effect in expected returns of 9.13% (equal weighted) and 6.65% (value-weighted). Unreported results indicate that the greater magnitude of the size effect in expected returns when compared to realized returns is driven by the second half of our sample period, consistent with the evidence in Hou and van Dijk (2008) that negative cash flow shocks to small firms and positive cash flow shocks to big firms caused the size effect in realized returns to be negligible over this period.

The value effect is considerably attenuated when we look at the equal-weighted average ICC estimates instead of realized returns. The equal-weighted return spread between the high and low BE/ME portfolios is 15.87% in realized returns, but only 8.68% in expected returns. The value-weighted average high-low BE/ME return spread is around 8% when measured in terms of either realized returns or the ICC.

The positive relation between leverage and realized returns carries over to our proxy for expected returns, but the *t*-statistics of the average 10-1 ICC spreads are much higher. Distress (measured using a hazard model) is positively related to both realized and expected returns. However, the realized return spread between the high and low distress decile portfolios is at best marginally statistically significant, while the return spread in expected returns is economically large and statistically significant. A distress-based 10-1 spread

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portfolio yields an equal-weighted ICC of 9.97% per annum (*t*-stat = 7.63) and a value-weighted ICC of 6.34% per annum (*t*-stat = 8.87). (We obtain similar results when we use Altman's Z-score or Ohlson's O-score to measure distress.)

High CAPEX firms have lower realized returns as well as lower expected returns than low CAPEX firms. The relation with expected returns is stronger in statistical terms, but slightly weaker in economic terms. The asset growth anomaly is much weaker in expected returns than in realized returns. A portfolio that is long in high asset growth firms and short in low asset growth firms earns an economically and statistically significant equal-weighted (value-weighted) average realized return of -13.79% (-8.28%) per annum. However, about three quarters of the realized return on this long-short portfolio observed over the sample period was unexpected. The asset growth effect in equal-weighted (value-weighted) average expected returns is only -3.29% (-1.85%).

The accrual anomaly is only present in realized returns. The economically and statistically equal-weighted (value-weighted) average 10-1 spread of -11.95% (-10.63%) in realized returns reduces to a -0.49% (-0.38%) ICC spread. Similarly, net operating assets are a negative predictor of realized returns, but bear little relation with expected returns. The equal-weighted (value-weighted) 10-1 realized return spread is -13.04% (-4.07%), while the corresponding ICC spread is only -2.84% (0.94%).

The negative relation between the dispersion in analysts' forecasts and stock returns is one the strongest anomalies we consider in this paper in terms of realized returns. Firms with low dispersion outperform firms with high dispersion by 14.54% per annum (equal-weighted) and 11.10% (value-weighted). Remarkably, the relation is reversed in expected returns. Firms with high dispersion have an equal-weighted (value-weighted) average ICC

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that is2.33% (3.80%) higher than firms with low dispersion. Both of these ICC spreads are statistically significant.

In sum, our results show that the empirical evaluation of asset pricing anomalies is sensitive to the choice of the expected return proxy. Our new measure of ex ante expected returns suggests that several of the well-known asset pricing anomalies represent sample-specific patterns in realized returns rather than reliable attributes of expected returns.

6. Conclusions

This paper proposes the implied cost of capital (ICC) based on earnings forecasts generated by a cross-sectional model as a new proxy for expected stock returns. We show that a parsimonious cross-sectional earnings model based on ex ante publicly available information generates earnings forecasts that outperform the consensus analyst forecasts in terms of forecast bias and earnings response coefficient. We find that the ICC estimates based on our model are a more reliable proxy for expected returns than those based on analysts' forecasts. We present evidence that this greater reliability can be attributed to the greater quality of the underlying earnings forecasts, Combined with the much greater cross-sectional and time-series coverage of the model-based ICC estimates, our approach produces expected return estimates that enable more reliable inferences from asset pricing tests and better decision making by corporate managers and investment professionals.

We demonstrate that our new expected return measure changes the inferences about key issues in the asset pricing literature by re-evaluating the equity premium and a number of anomalies based on the model-based ICC. We find that the value-weighted equity

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premium in ex ante expected returns is only around 1% per annum, consistent with the equity premium estimates derived by Mehra and Prescott (1985).

Our analysis also indicates that several of the well-known anomalies uncovered by empirical asset pricing studies do not reflect reliable patterns in expected returns. Most notably, the asset growth anomaly, the accrual anomaly, the net operating assets anomaly, and the analysts' forecast dispersion anomaly do not show up in ex ante expected returns as measured by our ICC estimates.

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References

Abarbanell, J., and V.L. Bernard, 1992, "Tests of Analysts' Overreaction/Underreaction to Earnings Information as an Explanation for Anomalous Stock Price Behavior," *Journal of Finance*, 47, 1181-1207.

Abarbanell, J., and B. Bushee, 1997, "Fundamental Analysis, Future Earnings, and Stock Prices," *Journal of Accounting Research*, 35, 1-24.

Ball, R., and P. Brown, 1968, "An Empirical Evaluation of Accounting Income Numbers," *Journal of Accounting Research*, 6, 158-178.

Blume, M.E., and I. Friend, 1973, "A New Look at the Capital Asset Pricing Model," *Journal of Finance*, 28, 19-33.

Brav, A., R. Lehavy, and R. Michaely, 2005, "Using Expectations to Test Asset Pricing Models," *Financial Management*, 34, 5-37.

Brown, L.D., 1993, "Earnings Forecasting Research: Its Implications for Capital Markets Research," *International Journal of Forecasting*, 9, 295-320.

Campbell, J.Y., J. Hilscher, and J. Szilagyi, 2008, "In Search of Distress Risk," *Journal of Finance,* 63, 2899-2939.

Campbell, J.Y., C. Polk, and T. Vuolteenaho, 2009, "Growth or Glamour? Fundamentals and Systematic Risk in Stock Returns," *Review of Financial Studies,* forthcoming.

Campbell, J.Y., and R.J. Shiller, 1988, "The Dividend-Price Ratio and Expectations of Future Dividends and Discount Factors," *Review of Financial Studies*, 1, 195-228.

Campello, M., L. Chen, and L. Zhang, 2008, "Expected Returns, Yield Spreads, and Asset Pricing Tests," *Review of Financial Studies*, 21, 1297-1338.

Chen, L., and X. Zhao, 2007, "What Drives Stock Price Movement?," working paper, Michigan State University / Kent State University.

Electronic copy available at: https://ssrn.com/abstract=1343516

Claus, J., and J. Thomas, 2001, "Equity Premia as Low as Three Percent? Evidence from Analysts' Earnings Forecasts for Domestic and International Stock Markets," *Journal of Finance,* 56, 1629-1666.

Cooper, M.J., H. Gulen, and M.J. Schill, 2008, "Asset Growth and the Cross-section of Stock Returns," *Journal of Finance*, forthcoming.

Diether, K.B., C.J. Malloy, and A. Scherbina, 2002, "Differences of Opinion and the Cross Section of Stock Returns," *Journal of Finance*, 57, 2113-2141.

Dugar, A., and S. Nathan, 1995, "The Effect of Investment Banking Relationships on Financial Analysts' Earnings Forecasts and Investment Recommendations," *Contemporary Accounting Research*, 12, 131-160.

Easton, P.D., 2008, "Estimating the Cost of Capital Implied by Market Prices and Accounting Data," monograph, University of Notre Dame.

Easton, P.D., and S.J. Monahan, 2005, "An Evaluation of Accounting-Based Measures of Expected Returns," *Accounting Review*, 80, 501-538.

Easton, P.D., and G.A. Sommers, 2007, "Effects of Analysts' Optimism on Estimates of the Expected Rate of Return Implied by Earnings Forecasts," *Journal of Accounting Research*, 45, 983-1015.

Elton, E.J., 1999, "Expected Return, Realized Return, and Asset Pricing Tests," *Journal of Finance*, 54, 1199-1220.

Fama, E.F., and K.R. French, 1993, "Common Risk Factors in the Returns on Stocks and Bonds," *Journal of Financial Economics*, 33, 3-65.

Fama, E.F., and K.R. French, 1997, "Industry Costs of Equity," *Journal of Financial Economics*, 43, 153-193.

Electronic copy available at: https://ssrn.com/abstract=1343516

Fama, E.F., and K.R. French, 2000, "Forecasting Profitability and Earnings," *Journal of Business*, 73, 161-175.

Fama, E.F., and K.R. French, 2002, "The Equity Premium," *Journal of Finance*, 57, 637-659.

Fama, E.F., and K.R. French, 2004, "New Lists: Fundamentals and Survival Rates," *Journal of Financial Economics*, 73, 229-269.

Fama, E.F., and K.R. French, 2006, "Profitability, Investment and Average Returns," *Journal of Financial Economics*, 82, 491-518.

Fama, E.F., and J.D. MacBeth, 1973, "Risk, Return, and Equilibrium: Empirical Tests," *Journal of Political Economy*, 81, 607-636.

Francis, J., I. Khurana, and R. Pereira, 2005, "Disclosure Incentives and Effects on Cost of Capital Around the World," *Accounting Review*, 80, 1125-1162.

Francis, J., P. Olsson, and D. Oswald, 2000, "Comparing the Accuracy and Explainability of Dividend, Free Cash Flow, and Abnormal Earnings Equity Value Estimates," *Journal of Accounting Research*, 38, 45-70.

Froot, K., and J.A. Frankel, 1989, "Forward Discount Bias: Is It an Exchange Risk Premium?," *Quarterly Journal of Economics*, 104, 139-161.

Gebhardt, W.R., C.M.C. Lee, and B. Swaminathan, 2001, "Toward an Implied Cost of Capital," *Journal of Accounting Research*, 39, 135-176.

Guay, W., S. Kothari, and S. Shu 2005, "Properties of Implied Cost of Capital Using Analysts' Forecasts," working paper, University of Pennsylvania.

Hail, L. and C. Leuz, 2006a, "International Differences in Cost of Capital: Do Legal Institutions and Securities Regulation Matter?," *Journal of Accounting Research*, 44 485-531.

Electronic copy available at: https://ssrn.com/abstract=1343516

Hail, L. and C. Leuz, 2006b, "Cost of Capital Effects and Changes in Growth Expectations Around U.S. Cross-Listings," working paper, University of Pennsylvania / University of Chicago.

Hirshleifer, D., K. Hou, S.H. Teoh, and Y. Zhang, 2008, "Do Investors Overvalue Firms With Bloated Balance Sheets?," working paper, University of California at Irvine / Ohio State University / Chinese University of Hong Kong.

Hong, H., and J. Stein, 1999, "A Unified Theory of Underreaction, Momentum Trading, and Overreaction in Asset Markets," *Journal of Finance*, 54, 2143-2184.

Hou, K., and D.T. Robinson, 2006, "Industry Concentration and Average Stock Returns," *Journal of Finance*, 61, 1927-1956.

Hou, K., and M.A. van Dijk, 2008, "Resurrecting the Size Effect: Firm Size, Profitability Shocks, and Expected Stock Returns," working paper, Ohio State University / Erasmus University.

Hribar, P., and N. Jenkins, 2004, "The Effect of Accounting Restatements on Earnings Revisions and the Estimated Cost of Capital," *Review of Accounting Studies*, 9, 337-356.

La Porta, R., 1996, "Expectations and the Cross-Section of Stock Returns," *Journal of Finance*, 51, 1715-1742.

Lev, B., and J.A. Ohlson, 1982, "Market-Based Empirical Research in Accounting: A Review, Interpretation, and Extension." *Journal of Accounting Research*, 21, 249-332.

Lin, H.-W., and M.F McNichols, 1998, "Underwriting Relationships, Analysts' Earnings Forecasts and Investment Recommendations," *Journal of Accounting and Economics* 25, 101-127.

Electronic copy available at: https://ssrn.com/abstract=1343516

Mehra, R., and E.C. Prescott, 1985, "The Equity Premium: A Puzzle," *Journal of Monetary Economics*, 15, 145-161.

O'Brien, P., 1988, "Analysts' Forecasts as Earnings Expectations," *Journal of Accounting and Economics*, 10, 53-83.

Pástor, L., M. Sinha, and B. Swaminathan, 2007, "Estimating the Intertemporal Risk-Return Tradeoff Using the Implied Cost of Capital," *Journal of Finance*, forthcoming.

Penman, S., and T. Sougiannis, 1998, "A Comparison of Dividend, Cash Flow, and Earnings Approaches to Equity Valuation," *Contemporary Accounting Research*, 15, 343-383.

Sharpe, W.F., 1978, "New Evidence on the Capital Asset Pricing Model: Discussion," *Journal of Finance*, 33, 917-920.

Sloan, R., 1996, "Do Stock Prices Fully Reflect Information in Accruals and Cash Flows About Future Earnings?," *The Accounting Review*, 71, 289-315.

Titman, S., K.C.J. Wei, and F. Xie, 2004, "Capital Investments and Stock Returns," *Journal of Financial and Quantitative Analysis*, 39, 677-700.

Vuolteenaho, T., 2002, "What Drives Firm-Level Stock Returns?," *Journal of Finance*, 57, 233-264.

Wiedman, C.I., 1996, "The Relevance of Characteristics of the Information Environment in the Selection of a Proxy for the Market's Expectations for Earnings: An Extension of Brown, Richardson, and Schwager [1987]," *Journal of Accounting Research*, 34, 313-324.

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Table 1: Cross-sectional earnings regressions, 1967-2005

This table reports the average slopes and their time-series *t*-statistics from annual pooled regressions of one-, two-, and three-year ahead earnings (E_{t+1}, E_{t+2}, and E_{t+3}, respectively) on variables that are hypothesized to capture differences in expected earnings across firms. Each pooled regression is estimated using the previous ten years (three year minimum) of data. V_t is the market value of the firm, A_t is the total assets, D_t is the dividend payment, DD_t is a dummy variable that equals 0 for dividend payers and 1 for non-payers, *Neg* E_t is a dummy variable that equals 1 for firms with negative earnings and 0 otherwise, and AC_t is the operating accruals.

LHS		Intercept	V_t	A_t	D_t	DD_t	E_t	*Neg* E_t	AC_t	Adj. R^2
E_{t+1}	Coefficient	2.1958	0.0140	-0.0079	0.2487	-2.8010	0.6839	-0.3305	-0.0510	0.87
	t-stat	*7.22*	*12.63*	*-6.89*	*8.11*	*-8.83*	*38.99*	*-0.79*	*-9.87*	
E_{t+2}	Coefficient	2.3349	0.0179	-0.0080	0.3939	-3.1478	0.5981	0.2372	-0.0571	0.81
	t-stat	*6.20*	*10.67*	*-4.74*	*9.33*	*-8.53*	*22.90*	*0.47*	*-6.19*	
E_{t+3}	Coefficient	2.5032	0.0225	-0.0088	0.4734	-3.4405	0.5398	0.8907	-0.0697	0.77
	t-stat	*7.20*	*10.65*	*-4.07*	*11.18*	*-9.85*	*18.80*	*2.68*	*-5.14*	

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Table 2: Summary statistics of earnings forecasts, 1967-2005

This table reports the year-by-year summary statistics of the one-, two-, and three-year ahead earnings forecasts based on the cross-sectional earnings model (Panel A) and those based on IBES analyst earnings forecasts (Panel B). $E_t[E_{t+1}]$, $E_t[E_{t+2}]$, and $E_t[E_{t+3}]$ are the value-weighted average one-, two-, and three-year ahead forecasted earnings (standardized by market capitalization for model-based forecasts and by price for IBES forecasts). N_{t+1}, N_{t+2}, and N_{t+3} are the number of firms with non-missing earnings forecasts. The IBES forecasts are the consensus forecasts as of June of each year. If the three-year ahead earnings forecast is missing from IBES, we use the consensus long-term growth estimate and the two-year ahead forecast to impute the three-year ahead forecast.

Year	N_{t+1}	$E_t[E_{t+1}]$	N_{t+2}	$E_t[E_{t+2}]$	N_{t+3}	$E_t[E_{t+3}]$	N_{t+1}	$E_t[E_{t+1}]$	N_{t+2}	$E_t[E_{t+2}]$	N_{t+3}	$E_t[E_{t+3}]$
	Panel A: Cross-sectional model						*Panel B: IBES*					
1967	1059	0.0545	1059	0.0618	1059	0.0691						
1968	1150	0.0633	1150	0.0669	1150	0.0730						
1969	1265	0.0858	1265	0.0959	1265	0.0988						
1970	1473	0.0550	1473	0.0593	1473	0.0641						
1971	1606	0.0550	1606	0.0557	1606	0.0590						
1972	1710	0.0666	1710	0.0685	1710	0.0702						
1973	1773	0.0970	1773	0.0977	1773	0.0998						
1974	2424	0.0983	2424	0.0996	2424	0.1019						
1975	2941	0.0859	2941	0.0897	2941	0.0962						
1976	3010	0.1034	3010	0.1046	3010	0.1125						
1977	2888	0.1122	2888	0.1172	2888	0.1236						
1978	2903	0.1213	2903	0.1255	2903	0.1335						
1979	2973	0.1309	2973	0.1377	2973	0.1434						
1980	2913	0.1167	2913	0.1268	2913	0.1372	1643	0.0826	1245	0.1024	4	0.0673
1981	2841	0.1517	2841	0.1652	2841	0.1809	1698	0.0978	1341	0.1162	1109	0.1335
1982	2884	0.0842	2884	0.0983	2884	0.1078	1842	0.0378	1492	0.0520	1307	0.0590
1983	3057	0.1036	3057	0.1172	3057	0.1319	2009	0.0574	1674	0.0724	1441	0.0783
1984	3024	0.0937	3024	0.1031	3024	0.1122	2082	0.0472	1717	0.0562	1475	0.0649
1985	3082	0.0664	3082	0.0727	3082	0.0809	2168	0.0344	1811	0.0420	1530	0.0462
1986	3156	0.0537	3156	0.0602	3156	0.0662	2223	0.0332	1907	0.0423	1628	0.0474
1987	3087	0.0736	3087	0.0805	3087	0.0858	2229	0.0469	1899	0.0542	1556	0.0605
1988	3122	0.0731	3122	0.0797	3122	0.0861	2395	0.0470	2084	0.0530	1693	0.0577
1989	3234	0.0705	3234	0.0784	3234	0.0863	2446	0.0409	2130	0.0499	1709	0.0576

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Table 2, continued

Year	N_{t+1}	$E_t[E_{t+1}]$	N_{t+2}	$E_t[E_{t+2}]$	N_{t+3}	$E_t[E_{t+3}]$	N_{t+1}	$E_t[E_{t+1}]$	N_{t+2}	$E_t[E_{t+2}]$	N_{t+3}	$E_t[E_{t+3}]$
	Panel A: Cross-sectional model						*Panel B: IBES*					
1990	3233	0.0618	3233	0.0703	3233	0.0760	2410	0.0339	2178	0.0450	1754	0.0540
1991	3168	0.0497	3168	0.0570	3168	0.0666	2450	0.0349	2221	0.0445	1849	0.0543
1992	3197	0.0462	3197	0.0527	3197	0.0579	2635	0.0372	2416	0.0510	2024	0.0578
1993	3353	0.0477	3353	0.0530	3353	0.0593	2910	0.0439	2701	0.0514	2266	0.0576
1994	3540	0.0477	3540	0.0503	3540	0.0535	3421	0.0405	3105	0.0479	2527	0.0579
1995	3796	0.0459	3796	0.0487	3796	0.0520	3767	0.0085	3408	0.0464	2808	0.0533
1996	3973	0.0418	3973	0.0453	3973	0.0488	3979	0.0343	3565	0.0463	3043	0.0533
1997	3965	0.0360	3965	0.0414	3965	0.0465	4189	0.0273	3803	0.0401	3287	0.0481
1998	3934	0.0270	3934	0.0329	3934	0.0405	4089	0.0296	3701	0.0393	3164	0.0452
1999	3797	0.0283	3797	0.0324	3797	0.0404	3817	0.0274	3293	0.0368	2839	0.0475
2000	3491	0.0346	3491	0.0406	3491	0.0491	3510	0.0221	3022	0.0348	2565	0.0557
2001	3434	0.0149	3434	0.0194	3434	0.0264	3498	0.0333	3223	0.0447	2742	0.0556
2002	3371	0.0182	3371	0.0209	3371	0.0232	3427	0.0448	3196	0.0539	2845	0.0622
2003	3222	0.0294	3222	0.0302	3222	0.0319	3530	0.0469	3371	0.0544	2949	0.0616
2004	3106	0.0373	3106	0.0375	3106	0.0398	3544	0.0546	3414	0.0619	2955	0.0684
2005	2890	0.0441	2890	0.0435	2890	0.0448	3299	0.0617	3196	0.0693	2755	0.0758

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Table 3: Average bias, accuracy, and earnings response coefficient of earnings forecasts, 1967-2005

Panels A, B, and C of this table report the time-series averages of the value-weighted average bias and accuracy as well as the cross-sectional earnings response coefficient (ERC) for the one-, two-, and three-year ahead earnings forecasts based on the cross-sectional earnings model and those based on IBES analysts' forecasts. $E_t[E_{t+1}]$, $E_t[E_{t+2}]$, and $E_t[E_{t+3}]$ are the one-, two-, and three-year ahead forecasted earnings. If the three-year ahead earnings forecast is missing from IBES, we use the consensus long-term growth estimate and the two-year ahead forecast to impute the three-year ahead forecast. Forecast bias is realized earnings minus earnings forecast (standardized by market capitalization for model-based forecasts and by price for IBES forecasts). Forecast accuracy is the absolute value of forecast bias. ERCs are estimated each year by running cross-sectional regressions of one-, two-, and three-year ahead realized returns on forecast biases (standardized by cross-sectional standard deviations of the biases) measured over the same horizon. Panel A reports the results for all the model-based forecasts available over the entire sample period (1967-2005). Panel B reports the results for the model-based forecasts for the 1980-2005 subperiod for firms for which both model-based and IBES forecasts are available (common sample). Panel C reports the results for the IBES forecasts for the common sample.

	$E_t[E_{t+1}]$	$E_t[E_{t+2}]$	$E_t[E_{t+3}]$
	Panel A: Cross-sectional model, full sample		
Bias	-0.0013	-0.0009	-0.0004
Accuracy	0.0277	0.0401	0.0507
ERC	0.0532	0.1727	0.2830
	Panel B: Cross-sectional model, common sample		
Bias	-0.0021	-0.0034	-0.0076
Accuracy	0.0272	0.0370	0.0476
ERC	0.0878	0.4166	0.9639
	Panel C: IBES, common sample		
Bias	-0.0060	-0.0167	-0.0189
Accuracy	0.0250	0.0294	0.0317
ERC	0.1066	0.1935	0.4127

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Table 4: Summary statistics of ICC estimates, 1968-2006

This table reports the year-by-year summary statistics of ICC estimates based on the cross-sectional earnings model (Panel A) and those based on IBES analysts' earnings forecasts (Panel B). The ICCs are estimated at the end of June of each year by solving for the discount rate required to equate the present value of forecasted earnings to the current market capitalization (Panel A) or equate the present value of the forecasted earnings per share to the current stock price (Panel B). We use a 12-year version of the residual income model which incorporates the earnings forecasts from either the cross-sectional earnings model or IBES for the next three years and require that the ROE reverts to the historical industry median between year 3 and year 12. Residual incomes after year 12 are assumed to be a perpetuity. N is the number of firms with a non-missing ICC estimate for a given year.

Year	N	Mean	STD	25%	Median	75%	N	Mean	STD	25%	Median	75%
	Panel A: Cross-sectional model						*Panel B: IBES*					
1968	1072	0.1543	0.1137	0.0899	0.1235	0.1760						
1969	1172	0.1397	0.0879	0.0918	0.1186	0.1580						
1970	1319	0.1722	0.1568	0.1144	0.1504	0.1916						
1971	1506	0.1215	0.0711	0.0813	0.1087	0.1416						
1972	1611	0.1098	0.0521	0.0788	0.1026	0.1322						
1973	1739	0.1393	0.0653	0.1073	0.1346	0.1612						
1974	1804	0.1650	0.0766	0.1253	0.1579	0.1900						
1975	2694	0.1680	0.1138	0.1176	0.1525	0.1927						
1976	2954	0.1780	0.1364	0.1176	0.1521	0.1975						
1977	2978	0.1819	0.1825	0.1201	0.1523	0.1985						
1978	2872	0.1712	0.1132	0.1174	0.1472	0.1864						
1979	2887	0.1759	0.1111	0.1216	0.1533	0.1957						
1980	2861	0.1845	0.1139	0.1257	0.1624	0.2062						
1981	2742	0.1733	0.1180	0.1150	0.1486	0.1924						
1982	2768	0.2021	0.1430	0.1381	0.1732	0.2190	889	0.1321	0.0610	0.0999	0.1268	0.1544
1983	2757	0.1426	0.1014	0.0925	0.1252	0.1653	1073	0.1050	0.0492	0.0759	0.1033	0.1271
1984	2823	0.1622	0.1195	0.1086	0.1382	0.1860	1297	0.1173	0.0902	0.0842	0.1092	0.1323
1985	2733	0.1532	0.1025	0.1018	0.1325	0.1726	1418	0.1098	0.0526	0.0851	0.1058	0.1275
1986	2826	0.1301	0.1013	0.0849	0.1124	0.1471	1477	0.1009	0.1076	0.0742	0.0935	0.1128
1987	2834	0.1235	0.1260	0.0787	0.1057	0.1409	1508	0.0968	0.0461	0.0733	0.0926	0.1134
1988	2842	0.1421	0.1299	0.0925	0.1172	0.1552	1488	0.1075	0.0646	0.0797	0.1017	0.1233
1989	2963	0.1648	0.2715	0.0944	0.1188	0.1583	1591	0.1031	0.0527	0.0776	0.0993	0.1191

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Table 4, continued

Year	N	Mean	STD	25%	Median	75%	N	Mean	STD	25%	Median	75%
	Panel A: Cross-sectional model						*Panel B: IBES*					
1990	2884	0.1883	0.5909	0.0891	0.1157	0.1567	1604	0.1044	0.1683	0.0722	0.0943	0.1171
1991	2669	0.1576	0.4898	0.0778	0.1013	0.1347	1630	0.1037	0.0681	0.0746	0.0975	0.1219
1992	2652	0.1072	0.1993	0.0672	0.0865	0.1122	1776	0.0980	0.0702	0.0715	0.0912	0.1128
1993	2775	0.0895	0.0715	0.0573	0.0794	0.1030	1917	0.0980	0.2181	0.0667	0.0871	0.1087
1994	3003	0.0941	0.0811	0.0636	0.0821	0.1059	2128	0.0999	0.1767	0.0679	0.0896	0.1104
1995	3171	0.0966	0.1045	0.0615	0.0820	0.1054	2399	0.0937	0.0493	0.0705	0.0909	0.1112
1996	3467	0.0941	0.0990	0.0604	0.0813	0.1038	2681	0.0972	0.1609	0.0682	0.0887	0.1082
1997	3494	0.1000	0.1749	0.0642	0.0818	0.1036	2864	0.0975	0.1992	0.0681	0.0867	0.1051
1998	3424	0.0920	0.1183	0.0639	0.0801	0.0994	3072	0.1063	0.6241	0.0662	0.0845	0.1044
1999	3292	0.0904	0.1213	0.0585	0.0799	0.1023	2912	0.1043	0.2518	0.0692	0.0910	0.1120
2000	3161	0.0983	0.1392	0.0574	0.0842	0.1163	2591	0.1033	0.0900	0.0666	0.0966	0.1231
2001	2831	0.0961	0.0962	0.0609	0.0818	0.1089	2380	0.1112	0.4805	0.0687	0.0891	0.1116
2002	2719	0.1093	0.3898	0.0547	0.0746	0.1016	2521	0.0973	0.1210	0.0728	0.0890	0.1051
2003	2643	0.0885	0.1363	0.0528	0.0733	0.0965	2692	0.1017	0.2159	0.0771	0.0932	0.1097
2004	2612	0.0726	0.0757	0.0477	0.0641	0.0820	2768	0.0867	0.0511	0.0709	0.0842	0.0975
2005	2593	0.0749	0.0568	0.0532	0.0691	0.0855	2730	0.1067	0.8727	0.0727	0.0854	0.0977
2006	1920	0.0730	0.1454	0.0504	0.0661	0.0838	2165	0.0897	0.0656	0.0749	0.0862	0.0986

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Table 5: Relation between realized returns and ICC estimates, 1968-2006

Panel A of this table reports the time-series averages of the cross-sectional correlations between realized returns and ICC estimates, as well as the average coefficients and their time-series *t*-statistics from annual Fama-MacBeth (1973) regression of realized returns on ICC estimates and proxies for cash flow news and discount rate news. Annual realized returns for year *t*+1, year *t*+2, and year *t*+3 (denoted by r_{t+1}, r_{t+2}, and r_{t+3}, respectively) are calculated by compounding monthly returns from July of year *t* to June of year *t*+1, from July of year *t*+1 to June of year *t*+2, and from July of year *t*+2 to June of year *t*+3, respectively. The ICCs are estimated at the end of June of year *t* by solving for the discount rate required to equate the present value of forecasted earnings to the current market capitalization (for cross-sectional model-based forecasts) or equate the present value of the forecasted earnings per share to the current stock price (for IBES forecasts). *Ncf* is the cash flow news proxy, which is taken from Easton and Monahan (2005) and represents the changes in the earnings forecasts for the three horizons we consider. *–Ndr* is the discount rate news proxy, which is the change in the ICC estimate for a given year multiplied by negative one. Both *Ncf* and *–Ndr* are orthogonalized with respect to ICC using cross-sectional regressions. Panel B of this table reports the time-series averages of the equal-weighted (EW) and value-weighted (VW) realized returns and their *t*-statistics over the next three years for decile portfolios sorted based on ICC measured at the end of June of year *t* as well as the realized return spread between Deciles 10 and 1 (10-1).

Panel A: Correlations and Fama-MacBeth cross-sectional regressions

			Fama-MacBeth regressions						Fama-MacBeth regressions				
		Corr.	Intercept	*ICC*	*Ncf*	*–Ndr*	Adj. R^2	Corr.	Intercept	*ICC*	*Ncf*	*–Ndr*	Adj. R^2
Dep. Var.													
				Cross-sectional model						*IBES*			
r_{t+1}	Coefficient	0.0550	0.1424	0.1318			0.01	-0.013	0.1714	-0.0320			0.01
	t-stat	*2.83*	*4.02*	*2.41*				*-0.54*	*3.79*	*-0.21*			
	Coefficient		0.1372	0.1741	0.2701	0.1021	0.10		0.1654	-0.0893	0.2311	0.2283	0.11
	t-stat		*3.60*	*2.31*	*9.95*	*7.14*			*4.29*	*-0.35*	*8.55*	*6.12*	
r_{t+2}	Coefficient	0.0512	0.1562	0.1195			0.00	-0.064	0.1981	-0.2585			0.00
	t-stat	*3.03*	*3.90*	*2.51*				*-2.77*	*4.67*	*-2.61*			
	Coefficient		0.1405	0.2193	0.2709	0.0851	0.09		0.2174	-0.5313	0.2609	0.2499	0.10
	t-stat		*3.73*	*3.24*	*10.38*	*8.11*			*6.54*	*-2.05*	*9.21*	*7.55*	
r_{t+3}	Coefficient	0.0429	0.1303	0.1575			0.00	-0.075	0.1640	-0.2064			0.01
	t-stat	*2.65*	*3.74*	*2.89*				*-3.80*	*5.94*	*-1.71*			
	Coefficient		0.1529	0.2483	0.2900	0.0936	0.09		0.2018	-0.3975	0.2503	0.2305	0.09
	t-stat		*4.43*	*3.07*	*10.52*	*7.63*			*7.05*	*-1.53*	*8.49*	*8.20*	

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Table 5, continued

Panel B: Returns on portfolios sorted on the ICC

Decile	r_{t+1}	*t-stat*	r_{t+2}	*t-stat*	r_{t+3}	*t-stat*	r_{t+1}	*t-stat*	r_{t+2}	*t-stat*	r_{t+3}	*t-stat*
	Cross-sectional model, EW returns						*IBES, EW returns*					
1	0.1047	*2.26*	0.0881	*2.19*	0.1228	*2.86*	0.1963	*3.93*	0.1834	*4.63*	0.2065	*6.04*
2	0.1305	*3.33*	0.1242	*3.48*	0.1464	*4.33*	0.1893	*4.37*	0.1679	*5.92*	0.1765	*7.93*
3	0.1367	*3.48*	0.1502	*4.01*	0.1636	*4.65*	0.1749	*4.28*	0.1456	*6.06*	0.1569	*7.27*
4	0.1481	*4.02*	0.1556	*4.30*	0.1680	*4.77*	0.1636	*4.24*	0.1390	*6.04*	0.1376	*5.92*
5	0.1477	*4.20*	0.1574	*4.17*	0.1678	*4.89*	0.1568	*4.24*	0.1301	*5.88*	0.1648	*7.11*
6	0.1578	*4.22*	0.1655	*4.42*	0.1801	*5.10*	0.1589	*5.21*	0.1304	*6.24*	0.1573	*6.74*
7	0.1777	*4.41*	0.1731	*4.56*	0.1799	*4.89*	0.1626	*4.62*	0.1343	*5.64*	0.1383	*5.31*
8	0.1777	*4.61*	0.1916	*4.84*	0.1917	*5.16*	0.1649	*4.70*	0.1173	*4.52*	0.1330	*5.73*
9	0.1864	*4.36*	0.1875	*4.47*	0.1974	*4.51*	0.1876	*4.94*	0.1387	*5.32*	0.1315	*4.47*
10	0.1935	*4.58*	0.1872	*4.37*	0.2129	*5.06*	0.1995	*4.53*	0.0997	*3.02*	0.0995	*2.98*
10-1	0.0888	*2.43*	0.0991	*3.33*	0.0901	*2.44*	0.0032	*0.08*	-0.0837	*-2.12*	-0.1070	*-2.82*
	Cross-sectional model, VW returns						*IBES, VW returns*					
1	0.0900	*2.57*	0.0866	*2.65*	0.1103	*3.65*	0.1781	*3.96*	0.1500	*4.19*	0.1766	*5.05*
2	0.1188	*3.69*	0.1066	*3.62*	0.1148	*4.36*	0.1605	*4.27*	0.1268	*4.11*	0.1327	*5.10*
3	0.1188	*3.91*	0.1216	*3.82*	0.1595	*5.63*	0.1320	*3.86*	0.1252	*4.28*	0.1401	*6.18*
4	0.1425	*4.65*	0.1485	*4.80*	0.1421	*5.44*	0.1386	*4.30*	0.1273	*5.23*	0.1362	*4.77*
5	0.1321	*4.61*	0.1499	*4.95*	0.1575	*5.57*	0.1342	*4.13*	0.1346	*5.46*	0.1578	*6.21*
6	0.1391	*5.09*	0.1525	*5.64*	0.1735	*6.97*	0.1469	*5.05*	0.1296	*5.57*	0.1265	*4.80*
7	0.1451	*4.63*	0.1382	*5.33*	0.1781	*6.32*	0.1512	*4.90*	0.1114	*4.09*	0.1419	*4.36*
8	0.1697	*5.25*	0.1769	*5.83*	0.1772	*6.02*	0.1538	*4.40*	0.1269	*4.49*	0.1237	*4.35*
9	0.1512	*4.32*	0.1768	*4.67*	0.1667	*4.35*	0.1485	*4.77*	0.1256	*4.53*	0.1198	*3.73*
10	0.1729	*4.11*	0.1622	*4.01*	0.1695	*4.44*	0.1324	*3.52*	0.0957	*2.65*	0.1076	*2.60*
10-1	0.0828	*2.50*	0.0757	*2.53*	0.0592	*2.25*	-0.0457	*-1.15*	-0.0544	*-1.57*	-0.0690	*-1.75*

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Table 6: Relation between realized returns and ICC estimates conditioning on forecast bias, accuracy, and ERC, 1968-2006

This table reports the time-series averages of the cross-sectional correlations between realized returns (Ret) and ICC estimates, as well as annual Fama-MacBeth (1973) regression coefficients of realized returns on ICC estimates, separately for firms with different levels of forecast bias, forecast accuracy, and earnings response coefficient (ERC). At the end of June of each year, we sort firms into tercile portfolios based on their forecast bias (Panel A), forecast accuracy (Panel B), or ERC (Panel C) for the next three years using the model-based earnings forecasts. We then calculate the correlations between the realized returns and the ICC estimates and estimate cross-sectional regressions of realized returns on the ICC estimates within each tercile. Annual realized returns are calculated by compounding monthly returns from July to June of next year. The ICCs are estimated at the end of June of each year by solving for the discount rate required to equate the present value of forecasted earnings to the current market capitalization. Forecast bias is realized earnings minus earnings forecast standardized by market capitalization. Forecast accuracy is the absolute value of forecast bias. We estimate firm-specific ERCs for each firm each year by dividing the firm's one-, two-, and three-year ahead realized returns by the forecast biases measured over the same horizon. Firms with negative ERCs are excluded from the analysis. Also reported in the tables are the time-series averages of the value-weighted bias, accuracy, ERC, realized return, and ICC of each tercile portfolio.

Panel A: Relation between realized returns and ICC estimates conditioning on forecast bias

	Tercile	Bias	Ret	ICC	Corr.	*t-stat*	FM	*t-stat*
	1	-0.1245	0.0685	0.1250	0.0864	*7.40*	0.3233	*5.94*
Bias $E_t[E_{t+1}]$	2	-0.0051	0.1063	0.0860	0.1270	*5.57*	1.1696	*4.36*
	3	0.0402	0.1761	0.0878	0.0745	*3.80*	0.6606	*2.55*
	1	-0.1986	-0.0315	0.1291	0.1477	*10.18*	0.5056	*8.04*
Bias $E_t[E_{t+2}]$	2	-0.0169	0.0660	0.0905	0.2009	*7.25*	1.7470	*7.45*
	3	0.0459	0.2395	0.0855	0.1198	*5.19*	1.2920	*3.68*
	1	-0.2352	-0.0120	0.1258	0.1074	*8.25*	0.3393	*6.22*
Bias $E_t[E_{t+3}]$	2	-0.0254	0.0529	0.0893	0.1723	*6.97*	1.4716	*6.65*
	3	0.0537	0.2244	0.0857	0.1161	*5.83*	1.2909	*5.50*

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Table 6, continued

Panel B: Relation between realized returns and ICC estimates conditioning on forecast accuracy

	Tercile	Accuracy	Ret	ICC	Corr.	*t-stat*	FM	*t-stat*
	1	0.0080	0.1176	0.0797	0.1051	*4.50*	1.0285	*3.55*
Accuracy $E_t[E_{t+1}]$	2	0.0379	0.1309	0.0953	0.0931	*4.32*	0.9416	*4.19*
	3	0.1909	0.1299	0.1246	0.0471	*3.47*	0.1351	*2.12*
	1	0.0136	0.1088	0.0801	0.1783	*6.16*	1.7552	*5.65*
Accuracy $E_t[E_{t+2}]$	2	0.0586	0.1358	0.0972	0.1185	*5.63*	1.1328	*5.35*
	3	0.2529	0.1328	0.1274	0.0420	*2.51*	0.0826	*1.08*
	1	0.0185	0.0964	0.0799	0.1579	*6.14*	1.4454	*6.32*
Accuracy $E_t[E_{t+3}]$	2	0.0748	0.1414	0.0969	0.0908	*4.19*	0.7463	*2.89*
	3	0.2978	0.1703	0.1237	0.0226	*1.52*	-0.0120	*-0.19*

Panel C: Relation between realized returns and ICC estimates conditioning on earnings response coefficient (ERC)

	Tercile	ERC	Ret	ICC	Corr.	*t-stat*	FM	*t-stat*
	1	1.80	-0.0004	0.0985	-0.0860	*-3.89*	-0.1898	*-3.49*
ERC $E_t[E_{t+1}]$	2	9.28	0.1014	0.0890	0.0415	*1.30*	0.6376	*2.21*
	3	176.46	0.2350	0.0799	0.1277	*3.98*	1.7139	*2.86*
	1	1.65	0.0077	0.0998	-0.0303	*-1.27*	-0.1559	*-1.93*
ERC $E_t[E_{t+2}]$	2	8.54	0.1098	0.0887	0.0761	*3.38*	1.2675	*4.14*
	3	160.26	0.1937	0.0804	0.1485	*5.05*	1.8752	*4.47*
	1	1.83	0.0433	0.0976	-0.0044	*-0.24*	-0.0690	*-0.91*
ERC $E_t[E_{t+3}]$	2	8.73	0.0976	0.0872	0.0642	*2.88*	1.0481	*3.19*
	3	143.53	0.1799	0.0821	0.1637	*7.02*	1.9526	*6.93*

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Table 7: Aggregate and industry realized and implied equity risk premia, 1968-2006

This table reports the time-series averages of the equal-weighted (Panel A) and value-weighted (Panel B) realized and implied risk premia for the aggregate market and for the 12 Fama-French industry portfolios. The industry definitions are downloaded from Ken French's Web site. Realized and implied risk premia are calculated by subtracting the risk-free rates from the realized returns and ICC estimates of the market and the 12 industry portfolios. Annual realized returns are calculated by compounding monthly returns from July to June of next year. The ICCs are estimated at the end of June of each year by solving for the discount rate required to equate the present value of forecasted earnings to the current market capitalization. We use two proxies for the risk-free rate. The first one (denoted "1") is the annualized 30-day T-Bill rate. The second proxy (denoted "2") is the annualized 10-year Treasury constant maturity rate.

Panel A: Equal-weighted risk premia

	Excess Ret1	*t-stat*	Excess Ret2	*t-stat*	Excess ICC1	*t-stat*	Excess ICC2	*t-stat*
Aggregate	0.0960	*2.52*	0.0804	*2.16*	0.0725	*15.92*	0.0569	*13.49*
Consumer NonDurables	0.0842	*2.23*	0.0686	*1.87*	0.0815	*19.40*	0.0658	*15.86*
Consumer Durables	0.0944	*1.88*	0.0788	*1.61*	0.0769	*17.69*	0.0613	*13.62*
Manufacturing	0.0914	*2.42*	0.0758	*2.06*	0.0765	*16.65*	0.0608	*14.86*
Energy	0.0916	*1.93*	0.0760	*1.60*	0.0409	*8.37*	0.0252	*6.85*
Chemicals	0.0923	*2.60*	0.0767	*2.22*	0.0708	*13.17*	0.0551	*11.38*
Business Equipment	0.1251	*2.21*	0.1095	*1.95*	0.0633	*12.93*	0.0477	*10.37*
Telecommunication	0.1126	*2.60*	0.0969	*2.29*	0.0564	*11.98*	0.0408	*10.14*
Utilities	0.0931	*3.40*	0.0774	*2.95*	0.0757	*19.08*	0.0600	*19.44*
Wholesale and Retail	0.0925	*2.04*	0.0768	*1.73*	0.0796	*16.29*	0.0640	*13.52*
Healthcare	0.1367	*2.81*	0.1210	*2.51*	0.0554	*10.98*	0.0398	*8.46*
Finance	0.0908	*2.30*	0.0752	*1.94*	0.0879	*9.47*	0.0723	*7.94*
Other	0.0833	*2.00*	0.0677	*1.66*	0.0767	*13.78*	0.0610	*11.79*

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Table 7, continued

Panel B: Value-weighted risk premia

	Excess Ret1	*t-stat*	Excess Ret2	*t-stat*	Excess ICC1	*t-stat*	Excess ICC2	*t-stat*
Aggregate	0.0599	*2.13*	0.0443	*1.63*	0.0260	*9.24*	0.0103	*6.26*
Consumer NonDurables	0.0763	*2.60*	0.0606	*2.13*	0.0222	*8.37*	0.0066	*2.91*
Consumer Durables	0.0476	*1.40*	0.0319	*0.98*	0.0395	*14.10*	0.0238	*9.05*
Manufacturing	0.0537	*1.82*	0.0380	*1.33*	0.0273	*9.66*	0.0117	*6.53*
Energy	0.0896	*2.78*	0.0739	*2.32*	0.0254	*6.11*	0.0097	*3.01*
Chemicals	0.0662	*2.18*	0.0505	*1.72*	0.0237	*8.20*	0.0081	*4.47*
Business Equipment	0.0619	*1.34*	0.0463	*1.02*	0.0115	*3.13*	-0.0042	*-1.90*
Telecommunication	0.0587	*1.90*	0.0431	*1.43*	0.0355	*10.09*	0.0198	*8.32*
Utilities	0.0681	*2.43*	0.0525	*1.94*	0.0557	*16.83*	0.0401	*18.59*
Wholesale and Retail	0.0732	*2.03*	0.0575	*1.64*	0.0266	*9.51*	0.0110	*4.03*
Healthcare	0.0780	*2.43*	0.0624	*1.96*	0.0137	*4.53*	-0.0020	*-0.95*
Finance	0.0779	*2.03*	0.0622	*1.66*	0.0295	*8.39*	0.0139	*4.07*
Other	0.0541	*1.53*	0.0385	*1.11*	0.0282	*7.53*	0.0126	*4.07*

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Table 8: ICC and firm characteristics, 1968-2006

This table reports the time-series averages of the equal-weighted (Panel A) and value-weighted (Panel B) realized returns (Ret) and ICC estimates for decile portfolios sorted based on various firm-level characteristics, as well as the 10-1 (High-Low) realized return and ICC spread and their *t*-statistics. Beta is estimated using a rolling 60-month (24 month minimum) window ending in June of each year. Size is market capitalization at the end of June of each year. BE/ME is book equity divided by market capitalization at previous year's fiscal year end. Leverage is book value of debt divided by book equity. Distress is the default probability estimated using the dynamic logit model as in Campbell et al. (2008). CAPEX is capital expenditure scaled by lagged total assets. Asset growth is the growth rate in total assets. Accruals is operating accruals scaled by lagged total assets. NOA is net operating assets standardized by lagged total assets. Analyst dispersion is the standard deviation of IBES analysts' earnings per share forecasts divided by the stock price as of June of each year.

Panel A: Equal-weighted portfolios

Characteristic		Low	2	3	4	5	6	7	8	9	High	10-1	*t-stat*
Beta	Ret	0.1633	0.1618	0.1720	0.1720	0.1677	0.1741	0.1437	0.1537	0.1388	0.1173	-0.0460	*-1.10*
	ICC	0.1681	0.1467	0.1384	0.1291	0.1301	0.1245	0.1232	0.1203	0.1206	0.1169	-0.0512	*-7.67*
Size	Ret	0.1719	0.1508	0.1437	0.1472	0.1510	0.1370	0.1416	0.1354	0.1314	0.1212	-0.0507	*-1.39*
	ICC	0.1749	0.1179	0.1083	0.1044	0.0999	0.0976	0.0946	0.0941	0.0919	0.0836	-0.0913	*-13.19*
BE/ME	Ret	0.0669	0.1087	0.1282	0.1387	0.1539	0.1680	0.1828	0.1851	0.2075	0.2257	0.1587	*5.38*
	ICC	0.1001	0.0985	0.1067	0.1118	0.1246	0.1306	0.1397	0.1500	0.1647	0.1868	0.0868	*15.66*
Leverage	Ret	0.1153	0.1419	0.1435	0.1715	0.1773	0.1752	0.1795	0.1708	0.1830	0.2033	0.0880	*2.44*
	ICC	0.0972	0.1051	0.1082	0.1115	0.1146	0.1207	0.1231	0.1279	0.1429	0.1548	0.0576	*13.68*
Distress	Ret	0.1298	0.1310	0.1474	0.1621	0.1709	0.1819	0.1743	0.1658	0.1840	0.2024	0.0726	*1.81*
	ICC	0.0882	0.0978	0.1076	0.1163	0.1238	0.1349	0.1449	0.1525	0.1652	0.1879	0.0997	*7.63*
CAPEX	Ret	0.1661	0.1695	0.1785	0.1647	0.1732	0.1630	0.1538	0.1625	0.1433	0.0984	-0.0677	*-2.70*
	ICC	0.1692	0.1445	0.1405	0.1338	0.1300	0.1263	0.1236	0.1235	0.1174	0.1119	-0.0573	*-8.36*

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Table 8, continued

Panel A: Equal-weighted portfolios

Characteristic		Low	2	3	4	5	6	7	8	9	High	10-1	*t-stat*
Asset Growth	Ret	0.1774	0.1695	0.1763	0.1648	0.1675	0.1564	0.1487	0.1285	0.1094	0.0395	-0.1379	*-6.66*
	ICC	0.1441	0.1382	0.1340	0.1286	0.1250	0.1222	0.1208	0.1228	0.1114	0.1112	-0.0329	*-7.42*
Accruals	Ret	0.1508	0.1682	0.1739	0.1343	0.1408	0.1585	0.1195	0.1207	0.0920	0.0313	-0.1195	*-7.30*
	ICC	0.1311	0.1344	0.1317	0.1333	0.1330	0.1288	0.1343	0.1279	0.1248	0.1262	-0.0049	*-0.65*
NOA	Ret	0.1864	0.1835	0.1926	0.1722	0.1807	0.1629	0.1622	0.1393	0.1327	0.0560	-0.1304	*-4.29*
	ICC	0.1451	0.1403	0.1409	0.1332	0.1348	0.1353	0.1339	0.1328	0.1276	0.1167	-0.0284	*-5.18*
Analyst Dispersion	Ret	0.2687	0.1994	0.1825	0.1621	0.1482	0.1498	0.1515	0.1373	0.1370	0.1233	-0.1454	*-4.48*
	ICC	0.0850	0.0830	0.0893	0.0930	0.0960	0.0995	0.1017	0.1052	0.1063	0.1083	0.0233	*7.08*

Panel B: Value-weighted portfolios

Characteristic		Low	2	3	4	5	6	7	8	9	High	10-1	*t-stat*
Beta	Ret	0.1196	0.1274	0.1324	0.1302	0.1360	0.1221	0.1076	0.1232	0.1225	0.1035	-0.0161	*-0.31*
	ICC	0.1003	0.0867	0.0836	0.0861	0.0889	0.0890	0.0898	0.0892	0.0926	0.0874	-0.0130	*-2.51*
Size	Ret	0.1600	0.1488	0.1433	0.1478	0.1498	0.1377	0.1422	0.1354	0.1307	0.1156	-0.0444	*-1.18*
	ICC	0.1458	0.1172	0.1081	0.1043	0.0998	0.0973	0.0945	0.0941	0.0915	0.0793	-0.0665	*-11.54*
BE/ME	Ret	0.1005	0.1202	0.1321	0.1230	0.1343	0.1445	0.1384	0.1778	0.1655	0.1850	0.0845	*2.43*
	ICC	0.0620	0.0737	0.0817	0.0875	0.0945	0.1049	0.1111	0.1170	0.1285	0.1433	0.0813	*17.97*
Leverage	Ret	0.1142	0.1213	0.1235	0.1118	0.1326	0.1318	0.1355	0.1251	0.1447	0.1441	0.0299	*0.73*
	ICC	0.0660	0.0736	0.0758	0.0813	0.0880	0.0945	0.0996	0.1073	0.1191	0.1145	0.0485	*15.80*

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Table 8, continued

Panel B: Value-weighted portfolios

Characteristic		Low	2	3	4	5	6	7	8	9	High	10-1	*t-stat*
Distress	Ret	0.1200	0.1366	0.1179	0.1315	0.1834	0.1470	0.1393	0.1417	0.1402	0.1870	0.0670	*1.39*
	ICC	0.0779	0.0901	0.0989	0.1031	0.1062	0.1120	0.1177	0.1183	0.1263	0.1413	0.0634	*8.87*
CAPEX	Ret	0.1381	0.1359	0.1245	0.1230	0.1492	0.1361	0.1146	0.1175	0.1138	0.0969	-0.0412	*-1.45*
	ICC	0.1039	0.0993	0.0947	0.0910	0.0925	0.0866	0.0856	0.0863	0.0836	0.0817	-0.0222	*-6.47*
Asset Growth	Ret	0.1584	0.1297	0.1391	0.1249	0.1264	0.1297	0.1190	0.1198	0.1049	0.0755	-0.0828	*-3.80*
	ICC	0.0989	0.0958	0.0934	0.0880	0.0838	0.0846	0.0807	0.0807	0.0792	0.0805	-0.0185	*-5.73*
Accruals	Ret	0.1165	0.1118	0.0884	0.1079	0.1180	0.1427	0.1094	0.0953	0.0794	0.0102	-0.1063	*-3.62*
	ICC	0.0854	0.0846	0.0827	0.0872	0.0835	0.0830	0.0839	0.0857	0.0837	0.0816	-0.0038	*-1.55*
NOA	Ret	0.1275	0.1356	0.1355	0.1192	0.1355	0.1281	0.1012	0.1221	0.1092	0.0867	-0.0407	*-1.50*
	ICC	0.0791	0.0826	0.0844	0.0833	0.0880	0.0922	0.0938	0.0939	0.0938	0.0886	0.0094	*2.97*
Analyst Dispersion	Ret	0.2063	0.1467	0.1522	0.0980	0.1170	0.1142	0.1097	0.1224	0.1311	0.0953	-0.1110	*-2.71*
	ICC	0.0697	0.0758	0.0795	0.0851	0.0884	0.0951	0.0949	0.0980	0.1012	0.1077	0.0380	*8.63*

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Figure 1: Time-series plot of the ICC and the implied equity risk premia, 1968-2006

The figure depicts the equal-weighted (Panel A) and value-weighted (Panel B) average model-based ICC as well as the implied risk premia relative to the annualized 30-day T-Bill rate (Excess ICC1) and the annualized 10-year Treasury constant maturity rate (Excess ICC2).

Panel A: Equal-weighted average ICC and implied equity risk premia

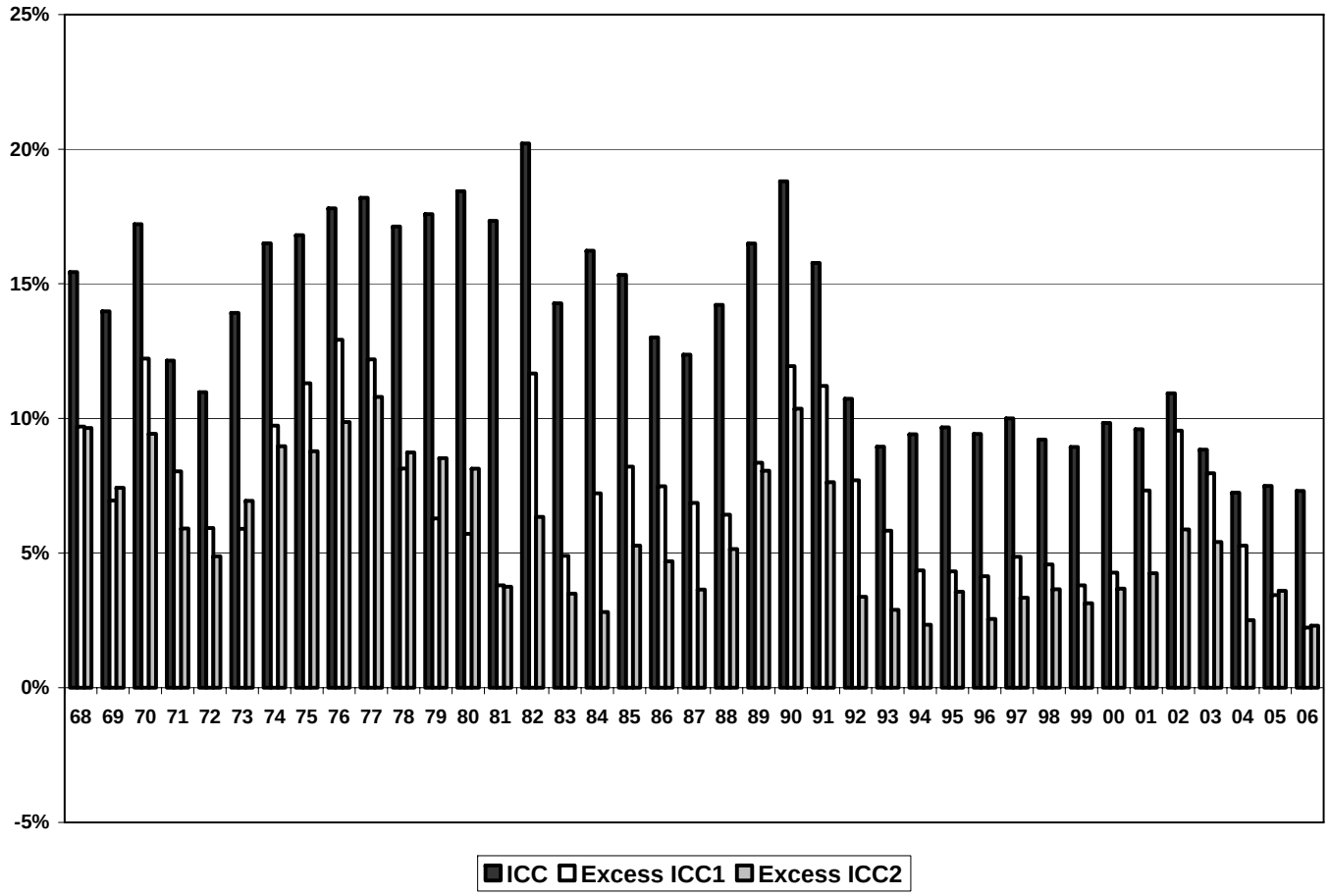

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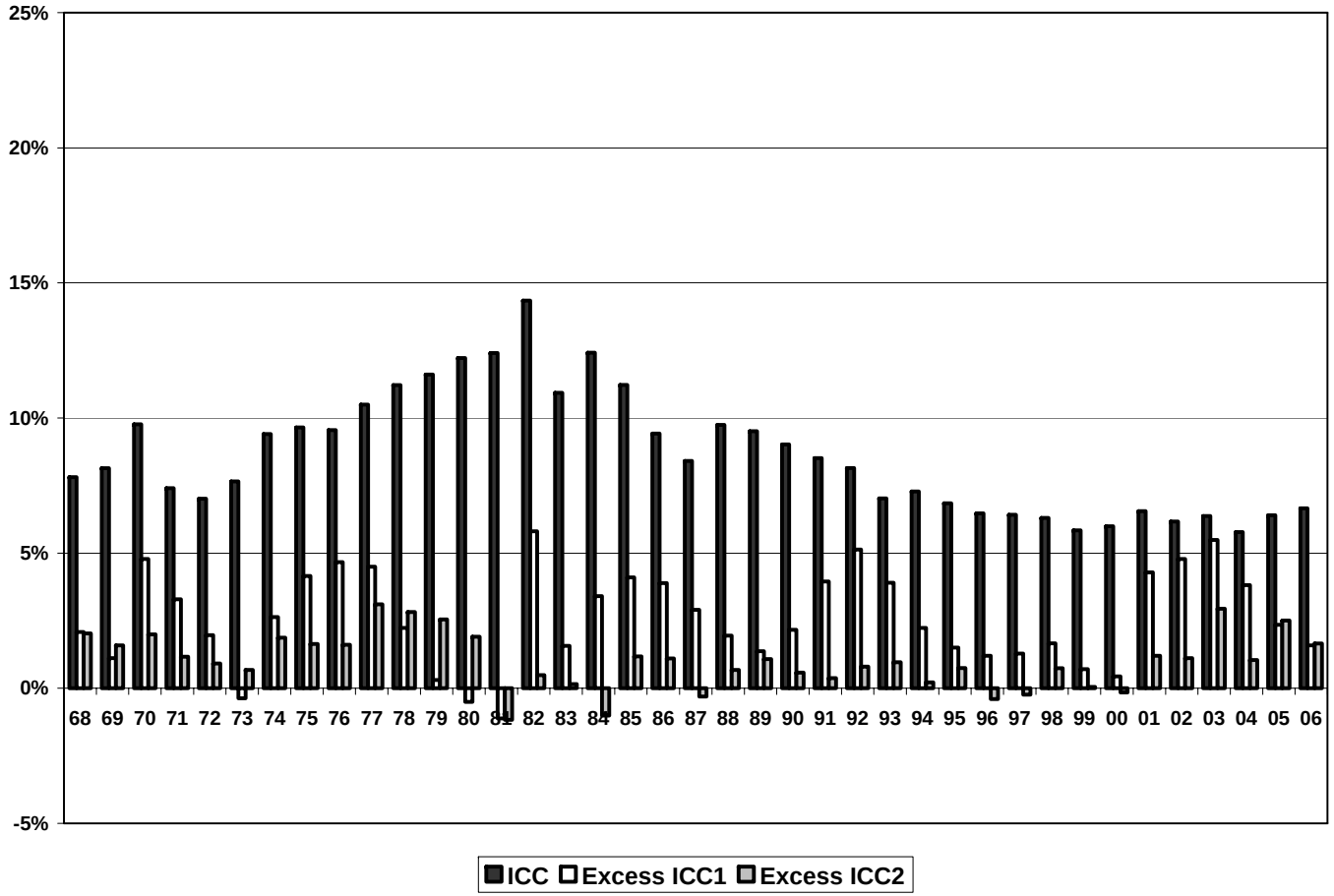
Figure 1, continued

Panel B: Value-weighted average ICC and implied equity risk premia



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